Alexander G. Cioffi, Ph.D.

Mass spectrometry-driven proteomics for drug discovery

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summary

Principal Scientist and founder of Cioffi Scientific with 9+ years advancing covalent and induced-proximity drug discovery through mass spectrometry-driven proteomics. Built and deployed critical-path assays across multiple therapeutic programs, spanning RapidFire-MS screens, cellular target engagement, and IND-enabling clinical sample analysis. Experienced leading direct reports and cross-functional collaborations, with publications in *Nature*, *Science*, and *Cell Chemical Biology*.

selected results

RapidFire-MS covalent screening
Built the high-throughput platform for covalent fragment hit ID; ran screens across 7 diverse targets.

3 TPD programs to cellular PoC
Carried three targeted protein degradation programs through key milestones, including cellular proof of concept.

Clinical PD biomarkers
Optimized sensitivity for p53 Y220C pharmacodynamic biomarkers in patient clinical samples.

experience

Founder · Cioffi Scientific — Oct 2026 – present

- Consult for small biotech companies on proteomics, chemoproteomics, and TPD assay strategy to advance programs through key milestones.
- Advise life-science AI teams on improving models for drug discovery.

Principal Scientist · Vulcan Bio — Mar 2025 – Sep 2026

- Spearheaded a chemoproteomic cellular target engagement platform (goDig) to drive hit-to-lead decisions across multiple programs.
- Engineered automated high-throughput cellular assays (BRET, AlphaLISA) to advance induced-proximity programs through key milestones.

Scientist I → Scientist II → Senior Scientist · Frontier Medicines — Aug 2019 – Feb 2025

- **Senior Scientist (2024–2025):** Conducted IND-enabling studies monitoring p53 Y220C target engagement in patient clinical samples; optimized sensitivity of the PD biomarker assays.
- **Scientist II (2022–2024):** Supervised two direct reports across three targeted protein degradation programs simultaneously, driving them from hit ID through lead generation and cellular proof of concept.
- Used diverse MS instrumentation (Q-ToF, Orbitrap Lumos/Eclipse, RapidFire) to develop critical-path assays, validate disease models, and inform key project decisions.
- **Scientist I (2019–2021):** Built and led RapidFire-MS high-throughput screening campaigns for covalent fragment hit ID across 7 diverse targets, establishing the platform for future programs.

Postdoctoral Researcher · UC Berkeley, Nomura Lab — Sep 2017 – Jul 2019

- Applied chemoproteomic covalent-ligand screening to discover a functional, ligandable cysteine within an intrinsically disordered region of MYC.

Graduate Researcher · University of Illinois Urbana-Champaign, Burke Lab — Sep 2011 – May 2017

- Guided a team of graduate students discovering small molecules that restore physiology in ion-transporter deficiencies; co-inventor on three patent applications.

education

Ph.D., Biochemistry · University of Illinois Urbana-Champaign · Advisor: Martin D. Burke — Sep 2011 – May 2017

B.S., Chemistry, Summa Cum Laude · Saint Louis University · Advisor: István Z. Kiss — Aug 2007 – May 2011

- First-author paper on electrochemical oscillations of nickel electrodissolution in a microchip flow cell.

skills

proteomics	Chemoproteomics, cellular target engagement, DIA / DDA / PRM, targeted protein degradation, covalent drug discovery, clinical sample analysis
instruments	Orbitrap Lumos/Eclipse, Q-ToF, RapidFire-MS
software	Skyline, DIA-NN, Proteome Discoverer, MaxQuant, Biopharma Finder, BioConfirm, Masspike; Python (pandas, NumPy, SciPy, matplotlib) and R
cell-based assays	BRET, AlphaLISA, automated high-throughput formats

publications

references available upon request

1. McMahon EJ, **Cioffi AG**, Visperas PR, Lin Y, Shaghafi M, Daczkowski CM, Hermann JC, Everley RA, Neve RM, Erlanson DA, Webster KR, Narayan V, Wang W. Structural basis for DCAF2 as a novel E3 ligase for PROTAC-mediated targeted protein degradation. *Structure* 2025, 33, 2020–2028.e7.
2. Boike L, **Cioffi AG**, Majewski FC, Co J, Henning NJ, Jones MD, Liu G, McKenna JM, Tallarico JA, Schirle M, Nomura DK. Discovery of a functional covalent ligand targeting an intrinsically disordered cysteine within MYC. *Cell Chem. Biol.* 2021, 28, 4–13.e17.
3. Muraglia KA, Chorghade RS, Kim BR, Tang XX, Shah VS, Grillo AS, Daniels PN, **Cioffi AG**, Karp PH, Zhu L, Welsh MJ, Burke MD. Small-molecule ion channels increase host defences in cystic fibrosis airway epithelia. *Nature* 2019, 567, 405–408.
4. Della Ripa LA, Petros ZA, **Cioffi AG**, Piehl DW, Courtney JM, Burke MD, Rienstra CM. Solid-state NMR of highly ^{13}C-enriched cholesterol in lipid bilayers. *Methods* 2018, 138–139, 47–53.
5. Grillo AS, SantaMaria AM, Kafina MD, **Cioffi AG**, Huston NC, Han M, Seo YA, Yien YY, Nardone C, Menon AV, Fan J, Svoboda DC, Anderson JB, Hong JD, Nicolau BG, Subedi K, Gewirth AA, Wessling-Resnick M, Kim J, Paw BH, Burke MD. Restored iron transport by a small molecule promotes absorption and hemoglobinization in animals. *Science* 2017, 356, 608–616.
6. Endo MM, **Cioffi AG**, Burke MD. Our path to less toxic amphotericins. *Synlett* 2016, 27, 337–354.
7. Davis SA, Della Ripa LA, Hu L, **Cioffi AG**, Pogorelov TV, Rienstra CM, Burke MD. C3-OH of amphotericin B plays an important role in ion conductance. *J. Am. Chem. Soc.* 2015, 137, 15102–15104.
8. **Cioffi AG**, Hou J, Grillo AS, Diaz KA, Burke MD. Restored physiology in protein-deficient yeast by a small molecule channel. *J. Am. Chem. Soc.* 2015, 137, 10096–10099.
9. Anderson TM, Clay MC, **Cioffi AG**, Diaz KA, Hisao GS, Tuttle MD, Nieuwkoop AJ, Comellas G, Maryum N, Wang S, Uno BE, Wildeman EL, Gonen T, Rienstra CM, Burke MD. Amphotericin forms an extramembranous and fungicidal sterol sponge. *Nat. Chem. Biol.* 2014, 10, 400–406.
10. **Cioffi AG**, Martin RS, Kiss IZ. Electrochemical oscillations of nickel electrodissolution in an epoxy-based microchip flow cell. *J. Electroanal. Chem.* 2011, 659, 92–100.

patent applications

1. Burke MD, Grillo AS, **Cioffi AG**, SantaMaria AM, Blair D, Blake A. Hinokitiol analogues, methods of preparing and pharmaceutical compositions thereof. PCT/US2019/027314, 2019.
2. Burke MD, Nardone C, Blake A, SantaMaria AM, Grillo AS, **Cioffi AG**, Lyu Y. Restoration of transmembrane copper transport. PCT/US2019/024340, 2019.
3. Burke MD, **Cioffi AG**, Diaz KA, Hou J, Grillo AS. Restoring physiology with small molecule mimics of missing proteins. PCT/US2015/058806, 2015.